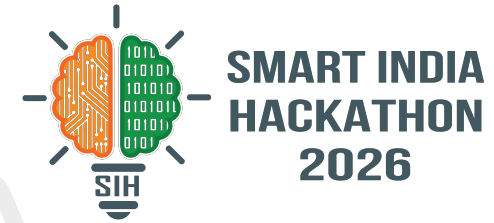
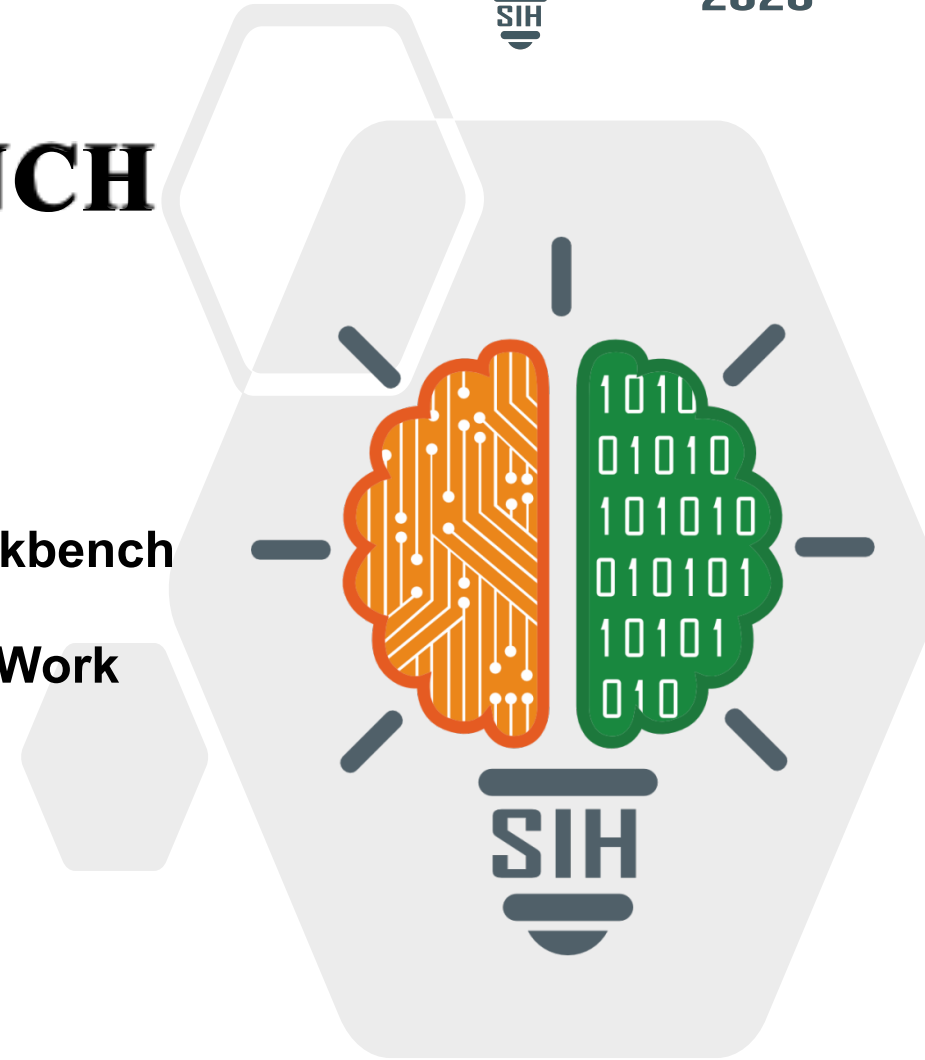


SMART INDIA HACKATHON 2026



SOVEREIGN AI WORKBENCH

- Problem Statement ID : 26117
- Problem Statement Title : Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work
- Theme : Smart Automation
- PS Category : Software
- Team ID : TBD
- Team Name : Tech Byte



A secure, self-hosted AI workbench that enables organizations to use powerful AI on confidential industrial data without sending data outside their infrastructure.



Your Data
Your Infrastructure



Open-Source &
Cost-Effective



Domain-Specific
Intelligence



Productivity
at Scale



Secure, Compliant
& Future-Ready



1. SECURE & PRIVATE

KEEP YOUR DATA IN YOUR CONTROL

- ✓ On-premise / air-gapped deployment
- ✓ Data stays inside the organization
- ✓ No external AI / cloud calls
- ✓ Role-based access control (RBAC)
- ✓ Secure access, audit logs & monitoring
- ✓ Designed for enterprise and government use



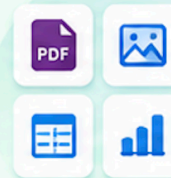
YOUR DATA | YOUR INFRASTRUCTURE | YOUR CONTROL



2. INTELLIGENT & MULTIMODAL

UNDERSTAND ANY TYPE OF DATA

- ✓ Multiple open-weight LLMs (e.g., Llama, Qwen)
- ✓ Automatic model selection based on task
- ✓ Understands PDFs, scans, images, drawings
- ✓ Local OCR + Vision + RAG
- ✓ Supports text, tables, charts and diagrams
- ✓ Works with domain-specific knowledge base
- ✓ Provides accurate, cited and verifiable answers



TEXT | IMAGES | TABLES | DRAWINGS | MULTIMODAL AI



3. AGENTIC & ACTION-ORIENTED

TURN INSIGHTS INTO ACTIONS

- ✓ Plans and executes multi-step tasks
- ✓ Uses local tools for files, code & spreadsheets
- ✓ Searches internal knowledge (RAG)
- ✓ Creates Word, PPT, Excel, reports & code
- ✓ Automates routine and complex workflows
- ✓ Human-in-the-loop for critical actions
- ✓ Scalable and customizable for different use cases



SEARCH | ANALYZE | ACT | AUTOMATE



Confidential &
Secure Operations



Higher
Productivity



Faster & Better
Decision Making



Reduced
Manual Effort

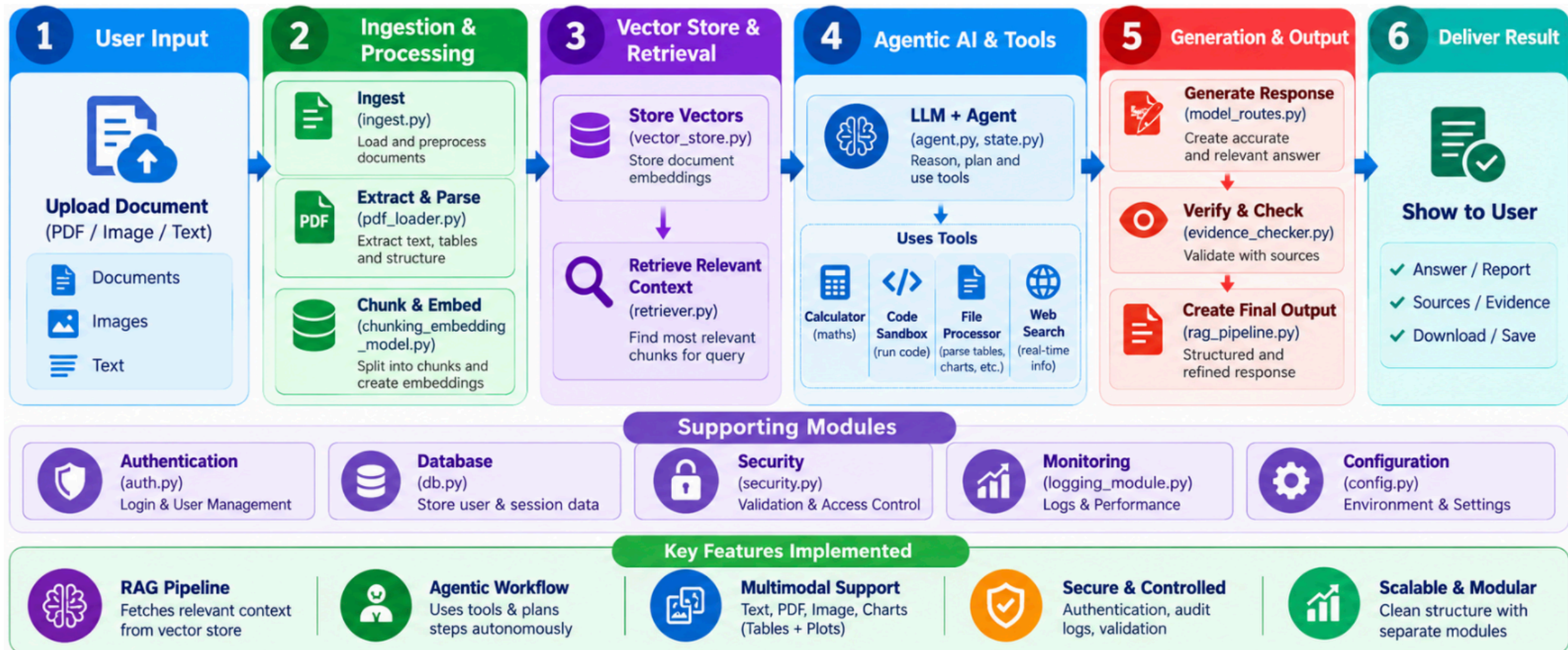


Cost-Effective
& Sustainable



Scalable for
Enterprise Use

BRING AI TO YOUR DATA – NOT YOUR DATA TO AI



CAN THIS WORK IN A REAL INDUSTRIAL ENVIRONMENT ?

Yes — with the right architecture, safeguards and domain adaptation, our solution is practical, scalable and ready for real-world use.

✓ REAL PROBLEMS
REAL SOLUTIONS

WHAT WE HAVE BUILT

KEY COMPONENTS OF OUR SOLUTION

-  **Local Open-Weight Models**
(Llama, LLaVA, Qwen)
Runs locally, no dependency on closed APIs ✓
-  **Agentic AI + Model Router**
(Task-based selection)
Automatically chooses the right model & tools ✓
-  **RAG + FAISS Knowledge Base**
(Company documents)
Uses internal data for accurate answers ✓
-  **OCR + Vision Processing**
(PDFs, images, drawings)
Understands text, tables, charts and diagrams ✓
-  **Local Tools & Automation**
(Files, Spreadsheets, Python)
Performs calculations, analysis and automation ✓
-  **Security & Access Control**
(JWT + RBAC + Audit)
Enterprise-grade security and data isolation ✓
-  **Python Sandbox**
(Safe code execution)
Executes analysis in a controlled environment ✓

KEY CHALLENGES

REAL-WORLD CONSTRAINTS

-  **GPU Limitations**
Limited compute resources for large models, especially in on-premise environments.
-  **Complex Documents**
Scanned PDFs, tables, drawings, handwritten content and multi-format data.
-  **AI Hallucination**
Incorrect or unsupported information.
-  **Sensitive Data**
Confidential industrial information must stay private and secure.
-  **Code Execution Safety**
Risk of malicious or harmful code execution.
-  **Integration & Scalability**
Working within existing IT systems and scaling for real users.

OUR MITIGATION STRATEGIES

HOW WE SOLVE IT

-  **Smaller / Task-Specific Models**
Optimized for local GPU/CPU and specific tasks (model routing, quantization).
-  **OCR + Vision + RAG**
Extract, understand and retrieve relevant information from complex documents.
-  **RAG + Evidence Check + Human Review**
Grounded responses with source references and verification.
-  **On-Premise + RBAC**
Keep data local with role-based access control, encryption and audit logs.
-  **Restricted Sandbox**
Safe and controlled code execution environment (with monitoring and limits).
-  **Modular & Scalable Architecture**
Containerized deployment, easy integration with enterprise systems, and support for future scaling.

TURNING SECURE AI INTO REAL-WORLD IMPACT

Secure Data | Faster Work | Smarter Decisions | Measurable Benefits

POTENTIAL IMPACT ON TARGET AUDIENCE



ENGINEERS & TECHNICAL TEAMS

WORK SMARTER, NOT HARDER

- ✓ Faster document & drawing analysis
- ✓ Instant SOP / manual search
- ✓ Quick access to standards & guidelines
- ✓ Reduce repetitive work
- ✓ Focus on design, analysis and innovation

Save Time
Increase Productivity





MANAGEMENT & DECISION MAKERS

FASTER DECISIONS, GREATER CONTROL

- ✓ Faster approvals and reviews
- ✓ AI-powered insights from documents
- ✓ Automated report generation
- ✓ Data-driven decision support
- ✓ Better compliance & governance
- ✓ Enhanced transparency

From Information to Action





ORGANIZATION / ENTERPRISE

SECURE • COMPLIANT • FUTURE READY

- ✓ Data sovereignty (keeps data on-premise)
- ✓ Local AI infrastructure
- ✓ Process automation & efficiency
- ✓ Regulatory compliance (RBAC, audit logs)
- ✓ Scalable for enterprise use
- ✓ Supports sustainable & paperless operations
- ✓ Knowledge retention and easy access

Secure Today
Stronger Tomorrow



KEY BENEFITS OF OUR SOLUTION



PRIVACY & SECURITY

Confidential data stays on-premise

TRUSTED & COMPLIANT



PRODUCTIVITY

Faster knowledge work & analysis

MORE WORK IN LESS TIME



COST EFFICIENCY

Less reliance on external tools

OPTIMIZED RESOURCES



AUTOMATION

Multi-step tasks handled by AI agents

REDUCED MANUAL EFFORT



KNOWLEDGE ACCESS

Organizational knowledge at your fingertips (RAG)

UNIFIED KNOWLEDGE HUB



ENVIRONMENT

Reduced paper-based workflow

GREENER OPERATIONS



RESEARCH FOUNDATION • DOCUMENTATION • SECURITY

Built on Trusted Research, Proven Frameworks and Industry Best Practices

1 RAG

Retrieval-Augmented Generation



- ✓ Ground responses with trusted data
- ✓ Reduces hallucinations
- ✓ Supports domain-specific knowledge
- ✓ Ideal for enterprise & government use

[↗ Research Paper](#)

2 ReAct

Reasoning + Acting with Tools



- ✓ Combines reasoning and action
- ✓ Uses external tools (search, calculator, DB)
- ✓ Improves factuality and reliability
- ✓ Effective for complex, multi-step tasks

[↗ Research Paper](#)

3 LangGraph

Agent Orchestration, Workflows & Routing



- ✓ Build complex agent workflows
- ✓ State management and memory
- ✓ Tool routing and human-in-the-loop
- ✓ Scalable and production-ready agents

[↗ Official Documentation](#)

4 Ollama

Local LLM Runtime for On-Premise AI



- ✓ Run open-source models locally
- ✓ Supports Llama, LLaVA, Qwen and more
- ✓ No dependency on closed APIs
- ✓ Ensures data privacy and control

[↗ Official Documentation](#)

5 FAISS

Vector Similarity Search & Retrieval



- ✓ High-performance vector search
- ✓ Enables fast and accurate retrieval
- ✓ Works with large-scale embeddings
- ✓ Widely adopted in RAG pipelines

[↗ Official Documentation](#)

6 NIST AI RMF

AI Risk Management Framework



- ✓ Guidelines for trustworthy AI
- ✓ Risk identification and mitigation
- ✓ Fairness, safety, privacy and security
- ✓ Helps in responsible & compliant deployment

[↗ Official Guidance](#)



Additional References
& Best Practices



Data Privacy &
Security Guidelines



Open-Source
Community Resources



Domain-Specific
Research Papers



Real-World
Case Studies



Industry Standards
& Compliance